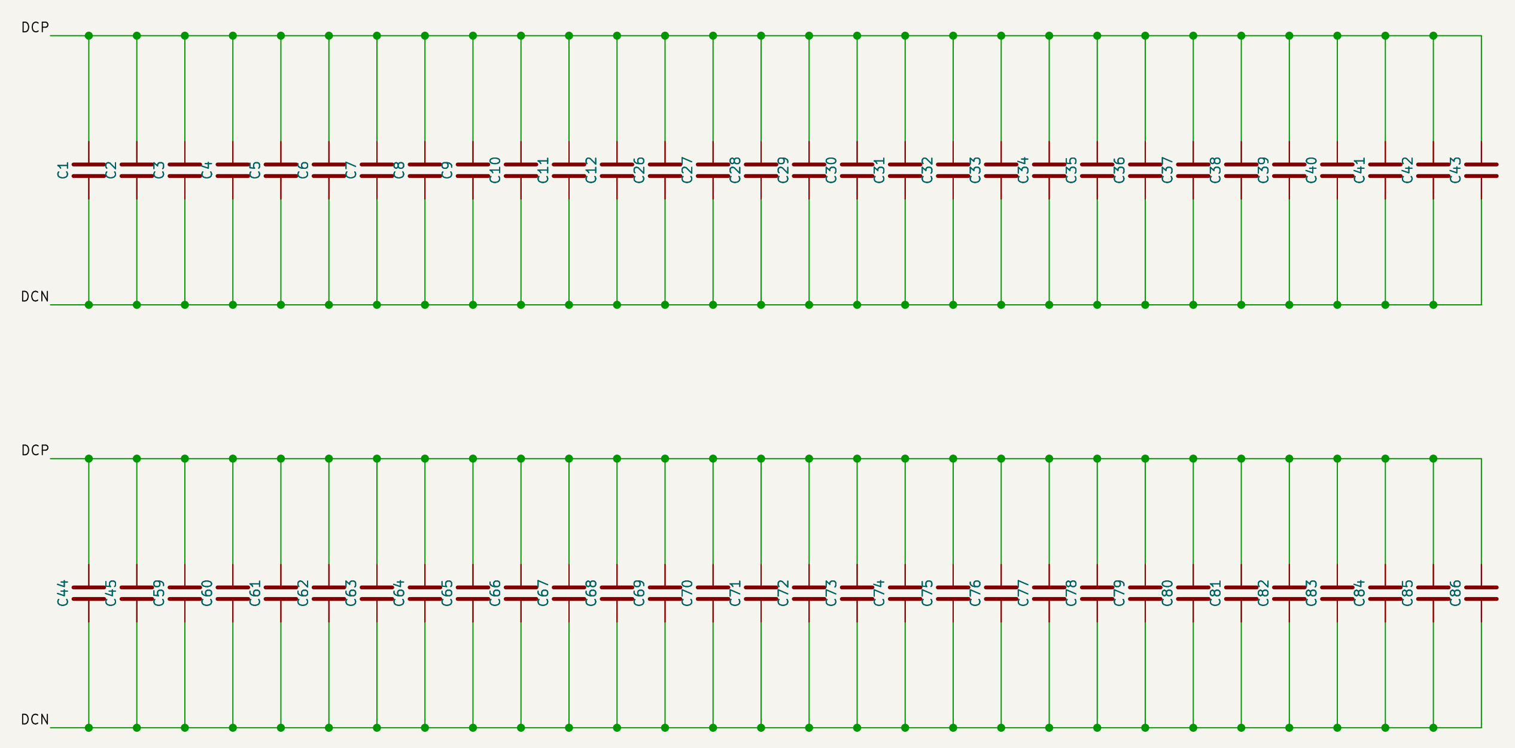
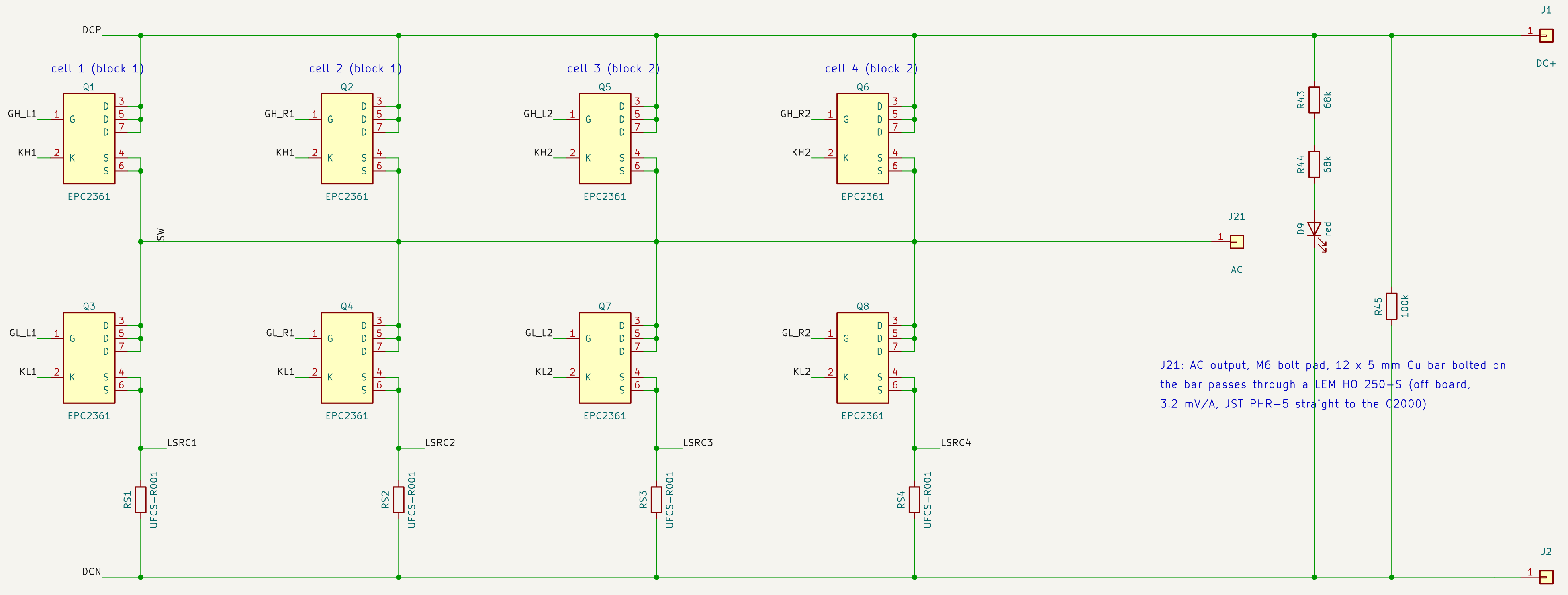
POWER STAGE AND DC LINK (block 1: Q1–Q4, block 2: Q5–Q8; one cell = HS + LS transistor + UFCS shunt in the LS source; AC of all four cells joined on the PCB)



DC link, 60 capacitors: 40 x 1uF 100V X7S, 20 x 22uF 100V X7S 2220 (1 uF 0805: bypass at the HS drains of each cell, top + bottom; 22 uF 2220: bank)



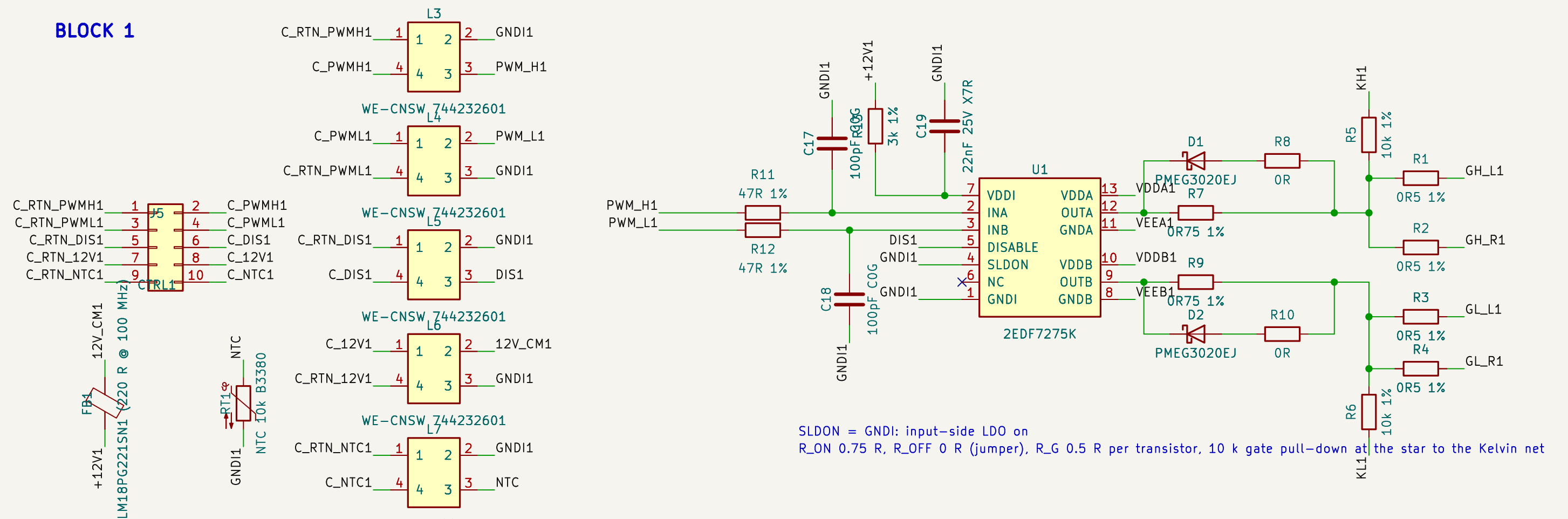
J21: AC output, M6 bolt pad, 12 x 5 mm Cu bar bolted on
the bar passes through a LEM HO 250-S (off board,
3.2 mV/A, JST PHR-5 straight to the C2000)

RS1–RS4: PMK UFCS-R001 1 mOhm coaxial shunt, IN = LS source (LSRCx), SHLD = DC−

DC link present LED, R45 100 k bleeder

GATE DRIVER BLOCKS: control header with common-mode chokes, 2EDF7275K input filters, R_ON / D + R_OFF, star, R_G per transistor (block 1 drives Q1–Q4, block 2 drives Q5–Q8)

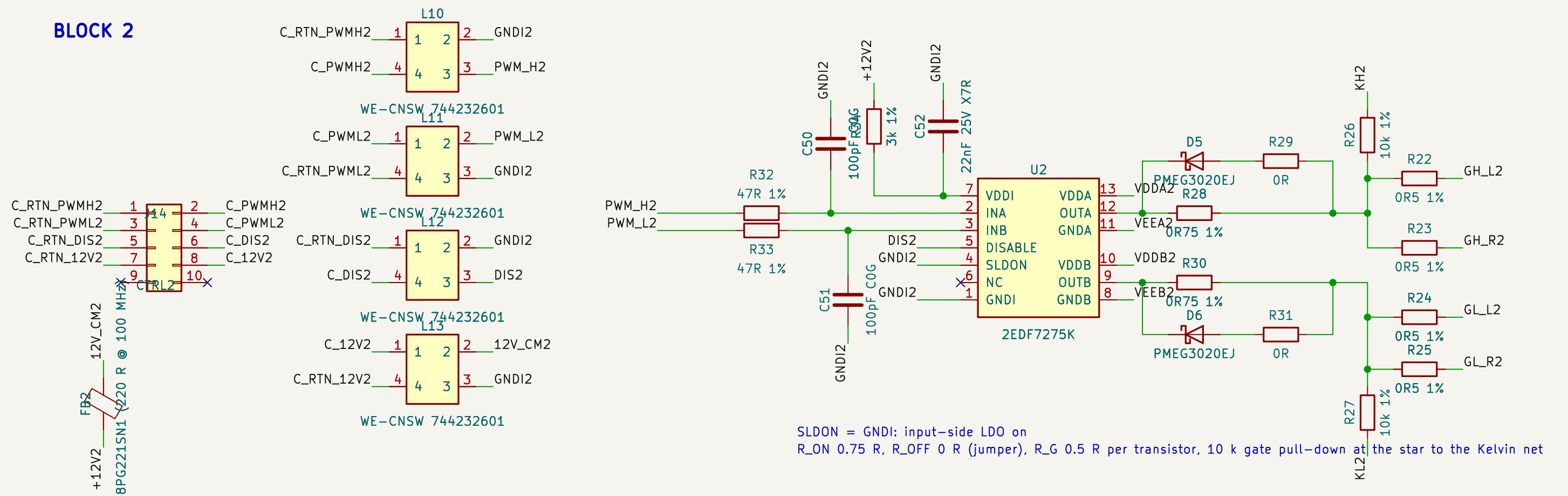
BLOCK 1



SLDON = GNDI: input-side LDO on
R_ON 0.75 R, R_OFF 0 R (jumper), R_G 0.5 R per transistor, 10 k gate pull-down at the star to the Kelvin net

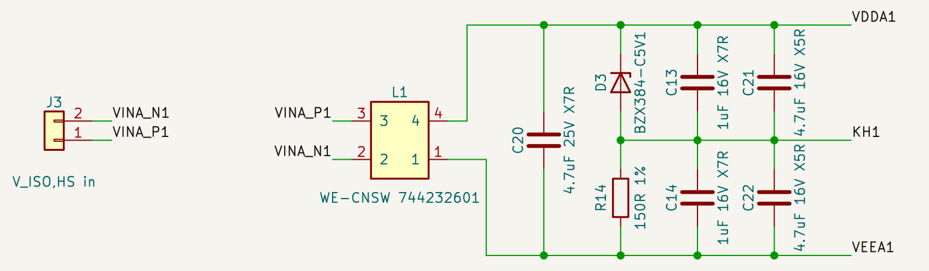
NTC on the control side (board temperature at the driver)

BLOCK 2

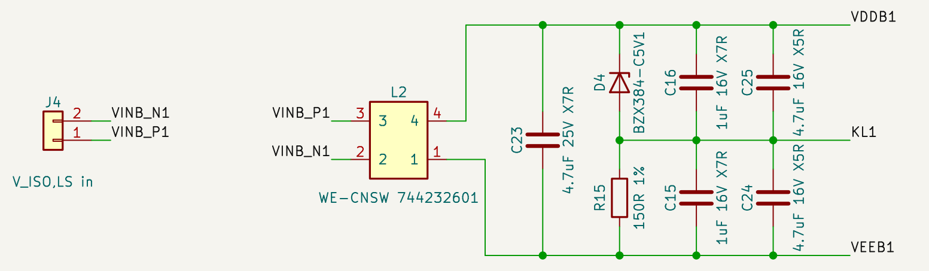


SLDON = GNDI: input-side LDO on
R_ON 0.75 R, R_OFF 0 R (jumper), R_G 0.5 R per transistor, 10 k gate pull-down at the star to the Kelvin net

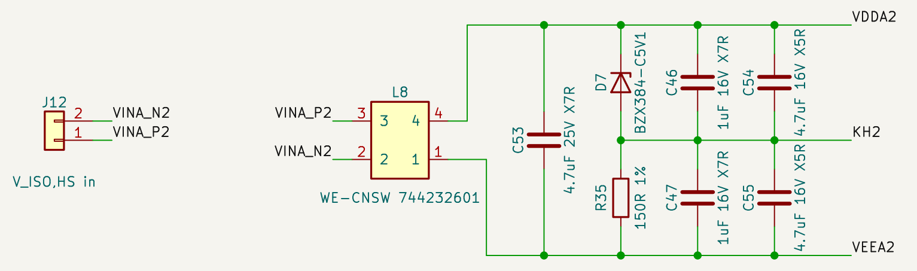
GATE SUPPLY ISLANDS: external isolated supply (one HS, one LS, split per block) -> WE-CNSW -> 5.1 V Zener VDD-S, 150 R bias S-VEE: V_GS,on = +5.1 V, V_GS,off = −(V_ISO − 5.1 V)



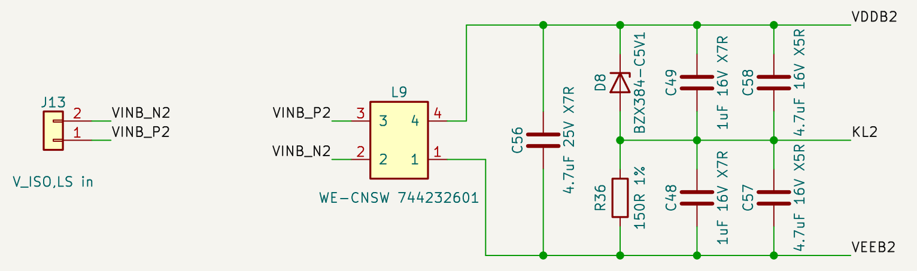
J3 / L1: HS block 1



J4 / L2: LS block 1

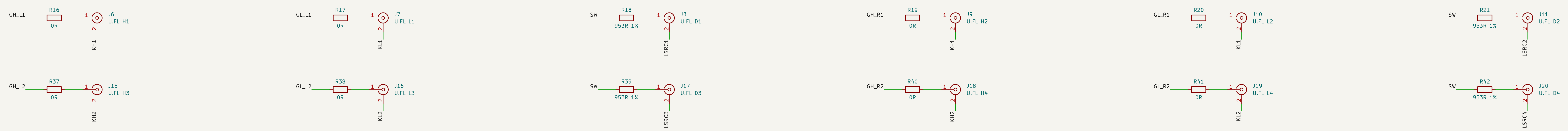


J12 / L8: HS block 2



J13 / L9: LS block 2

U.FL MEASUREMENT PORTS (bottom side, at each cell): V_GS HS / LS (0 R, shield = Kelvin), V_DS LS (953 R series = 20:1 into the 50 R scope input, shield = LS source LSRCx)



MOUNTING: H1–H4 board corners (insulated); no heatsink (double-pulse board)

Nets: DCP = DC+, DCN = DC−. SW = joined switch node of all four cells. KH1/KH2 and KL1/KL2 are the Kelvin source nets (pad 2 of each EPC2361); they meet the power source only inside the package.

LSRC1–LSRC4: LS power source of each cell = UFCS IN; the shunt SHLD side is DC−. The LS gate drive and the LS supply are referenced to the Kelvin net KLx, not to DC−.

Block suffix 1/2 on every driver-side net (GNDI1, VDDA2, …). Control side (CTRLx 2 x 5 header): PWM_H, PWM_L, DIS, 12 V and (block 1) NTC each with their own return C_RTN_* through one WE-CNSW common-mode choke per pair; 12 V then through a ferrite bead. The control side is a top-layer-only area with a 1200 V barrier (4 mm outer, 3 mm L2) to all power copper; see reports/pcb-rules/PCB_rules_and_checklist.md.

No short-circuit protection on this board. Closed-loop control from a TI TMS320F28379D over the CTRL headers.

generated by scripts/make_kicad_sch_revb.py; netlist identical to the PCB (check_sch_vs_pcb.py)
4 x EPC2361 per switch, 2 x 2EDF7275K, UFCS shunt per cell, no heatsink, external isolated supplies
KTH
Sheet: /
File: spb-bb-revb-dpt.kicad_sch

Title: SPB leg, Rev B, double-pulse board

Size: A1	Date: 2026-10-14	**Rev: B**
KiCad E.D.A. kicad-cli 7.0.10		Id: 1/1